

Yonghao Lee

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Education

2023–2027 **B.Sc. in Computer Science**, *The Hebrew University of Jerusalem*

Experience

Nov 2021–Jan 2026 **Cyber Threat Analyst**, *BrandShield Ltd.*

- Monitored and analyzed digital ecosystems to detect phishing sites, brand impersonations, and malicious infrastructure.
- Conducted deep-dive investigations into online fraud patterns and executed legal and technical takedown procedures.

Aug 2019–Jun 2021 **Sergeant**, *Israel Defense Forces (Armored Corps)*

- Received multiple outstanding soldier awards, including formal recognition from the Head of Southern Command (Aluf Pikud Darom) for exemplary performance.

Skills

Programming Python, C, C++, Java, Go, MATLAB, Unix, Web development

Tools & AI PyTorch, SpaCy, Git, LaTeX, Excel, Claude Code

Projects

2026 **Removing an Object's Light Footprint from a 3D Scene**

- Deleted an object from a **3D Gaussian Splatting** scene along with its shadow, mirror image, and color bleed: a video diffusion prior edits an orbit render under **SAM 2** masks from four clicks, then 3DGS is re-fit.
- Benchmarked on pixel-exact ground truth: **31.3 dB** PSNR on held-out views vs. 23.1 for Gaussian deletion.

2025 **Differentiable Rendering & Neural Radiance Fields**

- From-scratch differentiable rendering in **PyTorch**: SIREN neural fields, sphere tracing, volume rendering.
- Trained a **NeRF** (positional-encoded MLP) by volume rendering; re-renders the scene from novel viewpoints.

2025 **Multi-View 3D Reconstruction**

- From-scratch structure-from-motion in **Python/NumPy**: eight-point algorithm, essential-matrix pose recovery.
- Triangulated a 3D point cloud from two calibrated views, with image rectification.

Programs and Initiatives

Nov 2022–Jun 2023 **Project Lead & Social Entrepreneurship Fellow**, *Heseg Foundation*

- Led a team of 3 for a full year to initiate and execute a commercial food waste reduction program.
- Onboarded and managed partnerships with local restaurants, diners, and bakeries to adopt the initiative.
- Managed project operations, strategy, and waste control implementation as a Heseg Foundation Fellow.

Oct 2023–Present **Volunteer Mentor**, *Ach Gadol (Big Brother) for Lone Soldiers*

- Provide 1:1 personal guidance and crisis management to active-duty IDF Lone Soldiers.
- Help soldiers navigate complex military and civilian systems to secure housing, financial aid, and legal rights.

Oct 2023–Present **Volunteer**, *South African Zionist Federation*

- Assist new Olim (immigrants) with cultural adaptation, networking, and integration into Israeli society.
- Guide newcomers through local administrative processes, housing logistics, and community support resources.

Honors & Awards

2020 General's Outstanding Award, Southern Command (IDF)

2020 Outstanding Award, 460th Brigade (IDF)

2020 Outstanding Award, 9th Battalion (IDF)

Certifications & Languages

2023 **Responsive Web Design Developer Certification**, *freeCodeCamp*

Languages English (Native), Chinese (Native), Hebrew (Fluent), Japanese (Conversational)